VEHICLE RENTAL SYSTEM

Program:

```cpp
#include <iostream>
#include <string>
#include <vector>
using namespace std;

class Vehicle {
protected:
    int id;
    string brand, model;
    double rate;
    bool available;
public:
    Vehicle(int i, string b, string m, double r) {
        id = i;
        brand = b;
        model = m;
        rate = r;
        available = true;
    }

    virtual void displayVehicleInfo() {
        cout << "ID: " << id << endl;
        cout << "Brand: " << brand << endl;
        cout << "Model: " << model << endl;
        cout << "Rate/day: " << rate << endl;
        if(available==true){
            cout<<"Available: Yes"<<endl;
        }
        else {
            cout<<"Available: No"<<endl;
        }
    }

    void setAvailability(bool val) {
        available = val;
    }

    bool isAvailable() {
        return available;
    }

    int getID() {
        return id;
    }

    virtual double calculateRentalCost (int days) {
        return rate * days;
    }
};
```

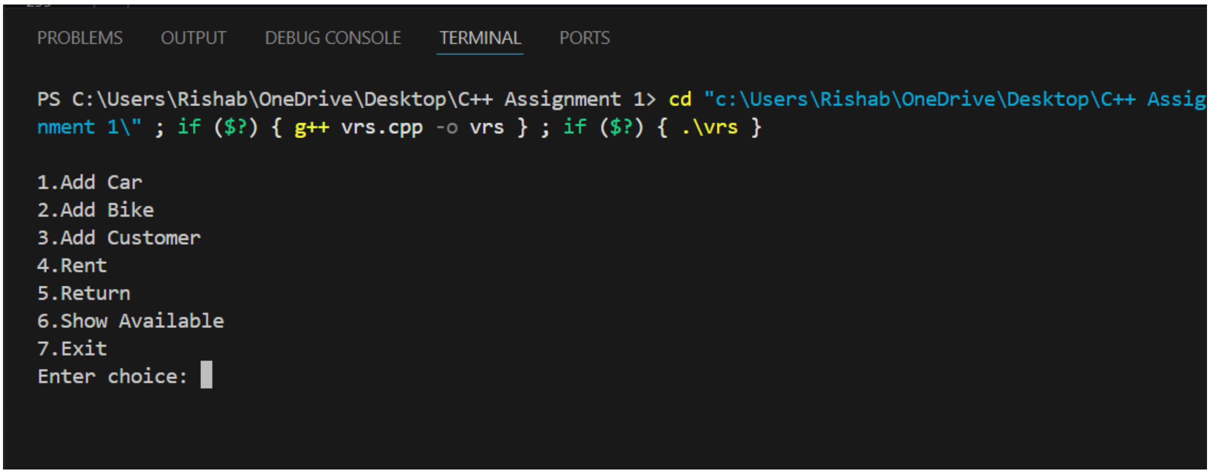
```cpp
class Car : public Vehicle {
    int seats;
    bool ac;
public:
    Car(int i, string b, string m, double r, int s, bool a)
        : Vehicle(i, b, m, r) {
        seats = s;
        ac = a;
    }

    double calculateRentalCost  (int days) override {
        double cost = rate * days;
        if (ac) cost += 200;
        return cost;
    }

    void displayVehicleInfo () override {
        Vehicle::displayVehicleInfo();
        cout << "Seats: " << seats << endl;

        if(ac==true)
        cout<<"AC: Yes"<<endl;
        else
        cout<<"AC: No"<<endl;
    }
};

class Bike : public Vehicle {
    string type;
    int cc;

public:
    Bike(int i, string b, string m, double r, string t, int c)
        : Vehicle(i, b, m, r) {
        type = t;
        cc = c;
    }
    double calculateRentalCost  (int days) override {
        double cost = rate * days;
        if (cc > 200) cost += 100;
        return cost;
    }
    void displayVehicleInfo () override {
        Vehicle::displayVehicleInfo();
        cout << "Type: " << type << endl;
        cout << "CC: " << cc << endl;
    }
};
```

```cpp
class Customer {
    int id;
    string name, license;
    vector<Vehicle*> rented;

public:
    Customer(int i, string n, string l) {
        id = i;
        name = n;
        license = l;
    }

    int getID() {
        return id;
    }

    void rentVehicle(Vehicle* v) {
        if (v->isAvailable()) {
            rented.push_back(v);
            v->setAvailability(false);
            cout << "Vehicle rented successfully" << endl;
        } else {
            cout << "Vehicle not available" << endl;
        }
    }

    void returnVehicle(int vid) {
        for (int i = 0; i < rented.size(); i++) {
            if (rented[i]->getID() == vid) {
                rented[i]->setAvailability(true);
                rented.erase(rented.begin() + i);
                cout << "Returned successfully" << endl;
                return;
            }
        }
        cout << "Vehicle not found" << endl;
    }

    void showRented() {
        cout << endl << "Rented Vehicles:" << endl;
        for (int i = 0; i < rented.size(); i++) {
            rented[i]->displayVehicleInfo();
            cout << "----------------" << endl;
        }
    }
};
```

```cpp
int main() {
    vector<Vehicle*> vehicles;
    vector<Customer> customers;

    int choice;

    do {
        cout << endl;
        cout << "1.Add Car" << endl;
        cout << "2.Add Bike" << endl;
        cout << "3.Add Customer" << endl;
        cout << "4.Rent" << endl;
        cout << "5.Return" << endl;
        cout << "6.Show Available" << endl;
        cout << "7.Exit" << endl;

        cout << "Enter choice: ";
        cin >> choice;

        if (choice == 1) {
            int id, seats;
            string b, m;
            double r;
            bool ac;

            cout << "Enter id brand model rate seats ac(1/0): ";
            cin >> id >> b >> m >> r >> seats >> ac;

            vehicles.push_back(new Car(id, b, m, r, seats, ac));
        }

        else if (choice == 2) {
            int id, cc;
            string b, m, t;
            double r;

            cout << "Enter id brand model rate type cc: ";
            cin >> id >> b >> m >> r >> t >> cc;

            vehicles.push_back(new Bike(id, b, m, r, t, cc));
        }
```

```
        else if (choice == 3) {
            int id;
            string name, lic;

            cout << "Enter id name license: ";
            cin >> id >> name >> lic;

            customers.push_back(Customer(id, name, lic));
        }
        else if (choice == 4) {
            int cid, vid, days;
            cout << "Enter customer id vehicle id days: ";
            cin >> cid >> vid >> days;

            for (int i = 0; i < customers.size(); i++) {
                if (customers[i].getID() == cid) {
                    for (int j = 0; j < vehicles.size(); j++) {
                        if (vehicles[j]->getID() == vid && vehicles[j]->isAvailable()) {
                            customers[i].rentVehicle(vehicles[j]);
                            cout << "Cost: " << vehicles[j]->calculateRentalCost(days) << endl;
                        }
                    }
                }
            }
        }
        else if (choice == 5) {
            int cid, vid;
            cout << "Enter customer id and vehicle id: ";
            cin >> cid >> vid;
            for (int i = 0; i < customers.size(); i++) {
                if (customers[i].getID() == cid) {
                    customers[i].returnVehicle(vid);
                }
            }
        }
        else if (choice == 6) {
            cout << endl << "Available Vehicles:" << endl;
            for (int i = 0; i < vehicles.size(); i++) {
                if (vehicles[i]->isAvailable()) {
                    vehicles[i]->displayVehicleInfo();
                    cout << "---------------" << endl;
                }
            }
        }

    } while (choice != 7);

    return 0;
}
```

OUTPUTS:



```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\Rishab\OneDrive\Desktop\C++ Assignment 1> cd "c:\Users\Rishab\OneDrive\Desktop\C++ Assig
nment 1\" ; if ($?) { g++ vrs.cpp -o vrs } ; if ($?) { .\vrs }

1.Add Car
2.Add Bike
3.Add Customer
4.Rent
5.Return
6.Show Available
7.Exit
Enter choice:
```

```
Enter choice: 1
Enter id brand model rate seats ac(1/0): 345
Tata
500
4
1
1

1.Add Car
2.Add Bike
3.Add Customer
4.Rent
5.Return
6.Show Available
7.Exit
Enter choice: 2
Enter id brand model rate type cc: 555
Royal
tyul
1000
200
200
```

```
Enter choice: 3
Enter id name license: 444
Rishab
12348

1.Add Car
2.Add Bike
3.Add Customer
4.Rent
5.Return
6.Show Available
7.Exit
Enter choice: 4
Enter customer id vehicle id days: 444
555
10
Vehicle rented successfully
Cost: 10000
```

```
Enter choice: 5
Enter customer id and vehicle id: 444
555
Returned successfully

1.Add Car
2.Add Bike
3.Add Customer
4.Rent
5.Return
6.Show Available
7.Exit
Enter choice: 6

Available Vehicles:
ID: 345
Brand: Tata
Model: 500
Rate/day: 4
Available: Yes
Seats: 1
AC: Yes
----------------
ID: 555
Brand: Royal
Model: tyul
Rate/day: 1000
Available: Yes
Type: 200
CC: 200
----------------
```